Angle Relationships: Mini-Lessons

4.1 Angle pairs NC.8.G.5

Angles come in pairs that follow rules. **Complementary** angles add to **90°**. **Supplementary** angles add to **180°**. **Vertical** angles (across from each other where two lines cross) are **equal**.

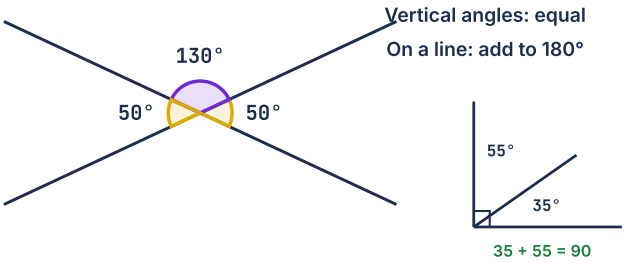
Complementary
two angles that add to 90° (a corner)

Supplementary
two angles that add to 180° (a straight line)

Vertical angles
across from each other at a crossing: equal

Adjacent
next to each other, sharing a side

Watch out: Mixing up 90 and 180?
Complementary = **C**orner = 90°.
Supplementary = **S**traight = 180°.



Walk through: `Two angles are supplementary. One is 110°. Find the other.`

1. Supplementary angles add to...
   180° → ? + 110 = 180
2. How do we find the missing angle?
   Subtract: 180 − 110 → 180 − 110 = 70
3. The other angle is...
   70° → The other angle = 70°

Check your understanding

1 Score 3 NC.8.G.5

Find the **complement** of 62°.

Answer: **28°**

1 `90 - 62 = 28°`

2 Score 3 NC.8.G.5

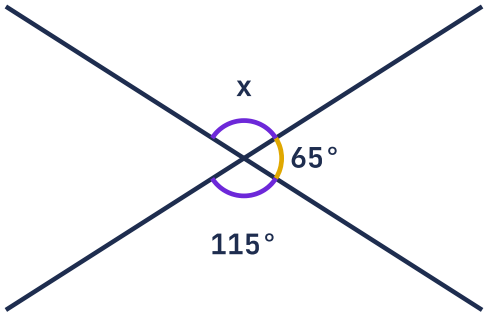
Find the **supplement** of 145°.

Answer: **35°**

1 `180 - 145 = 35°`

3 Score 3 NC.8.G.5

Two lines cross. Find **x**.



Answer: **115°**

1 `x and 115° are vertical angles → equal`

2 `x = 115°`

4.2 Parallel lines cut by a transversal NC.8.G.5

When a transversal cuts two **parallel** lines, every angle is either **equal** to a given angle or **adds with it to 180°**. Equal pairs: **corresponding, alternate interior, alternate exterior, vertical**. Same-side interior angles add to **180°**.

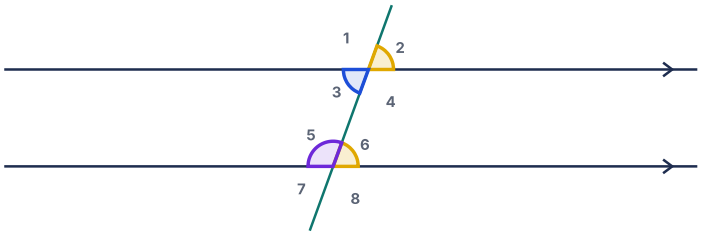
Transversal
a line that crosses two other lines

Corresponding
same corner at each crossing: equal

Alternate interior
between the lines, opposite sides: equal

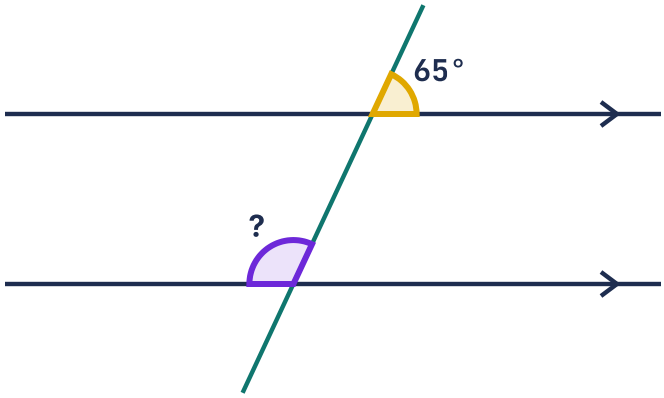
Same-side interior
between the lines, same side: add to 180°

Watch out: Look at the picture: a small-looking angle can only equal another small-looking angle.



small = 70°, big = 110°

Walk through: `Lines are parallel. ∠2 = 65°. Find ∠5.`

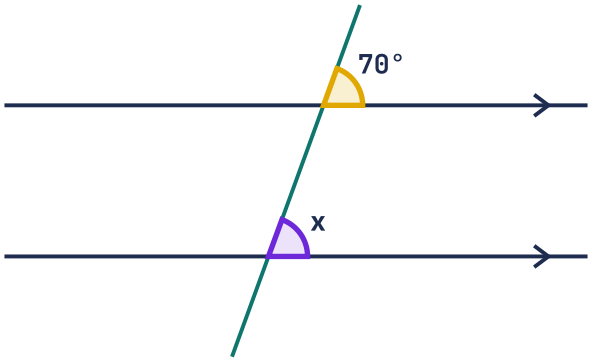


1. Is ∠5 small-looking (acute) or big-looking (obtuse)?
   Big (obtuse) → ∠2 is small, ∠5 is big
2. So ∠5 and ∠2 are...
   Supplementary: they add to 180° → ∠5 + 65° = 180°
3. Find ∠5.
   115° → ∠5 = 180 − 65 = 115°

Check your understanding

1 Score 3 NC.8.G.5

The lines are parallel. Find **x**.

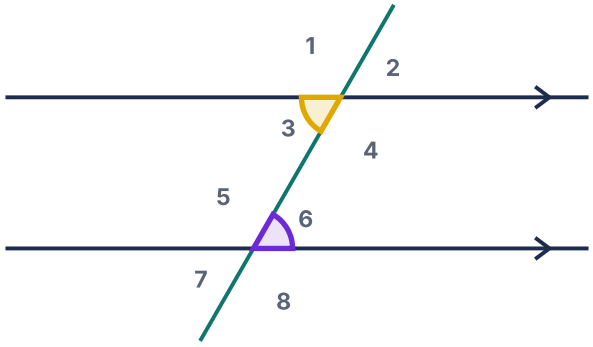


Answer: **70°**

1. `corresponding angles → equal`
2. `x = 70°`

2 Score 3 NC.8.G.5

What kind of pair are ∠3 and ∠6?



Ⓐ Corresponding angles

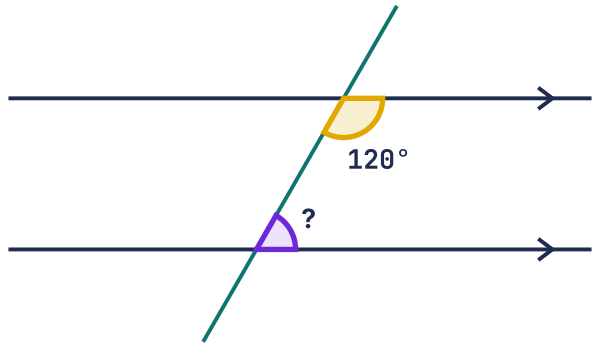
Ⓑ **Alternate interior angles**

Ⓒ Vertical angles

Ⓓ Same-side interior angles

3 Score 3 NC.8.G.5

The lines are parallel. ∠4 = 120°. Find ∠6 (a same-side interior angle).



Answer: **60°**

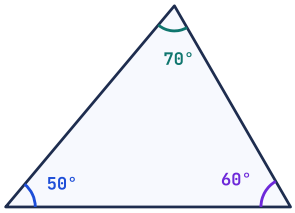
1 `180 - 120 = 60°`

4.3 Angles in a triangle NC.8.G.5

The three inside angles of **any** triangle add to **180°**. To find a missing angle: add the two you know, then subtract from 180.

Interior angle
an angle inside the triangle

Right triangle
a triangle with one 90° angle (a square corner)

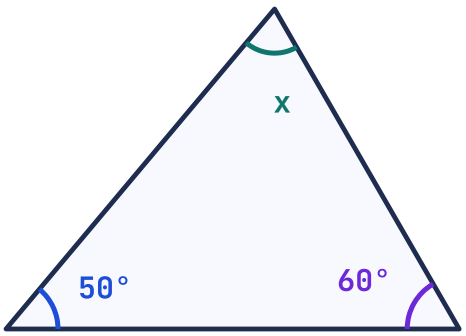


`50 + 70 + 60 = 180`



Watch out: Use 180°, not 360°. 360° is a full circle (or a four-sided shape).

Walk through: `A triangle has angles 50° and 60°. Find the third angle.`

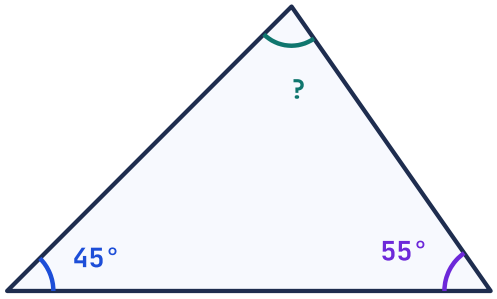


1. The three angles of a triangle add to...
   180° → 50 + 60 + x = 180
2. Add the angles you know.
   50 + 60 = 110 → 110 + x = 180
3. Find x.
   180 − 110 = 70° → x = 70°

Check your understanding

1 Score 3 NC.8.G.5

A triangle has angles 45° and 55°. Find the third angle.

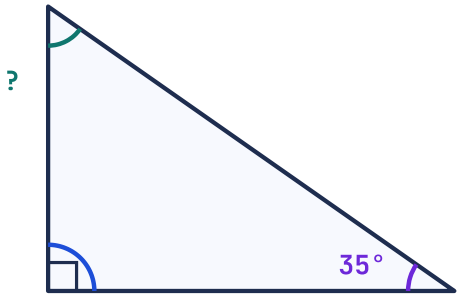


Answer: **80°**

1. `45 + 55 = 100`
2. `180 - 100 = 80°`

2 Score 3 NC.8.G.5

A right triangle has one angle of 35°. Find the other small angle.



Answer: **55°**

1. `90 + 35 = 125`
2. `180 - 125 = 55°`

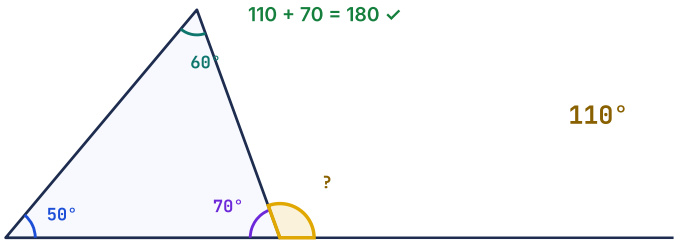
4.4 The exterior angle of a triangle NC.8.G.5

An **exterior angle** is outside the triangle, made by stretching one side. It **equals the two far inside angles added together**.

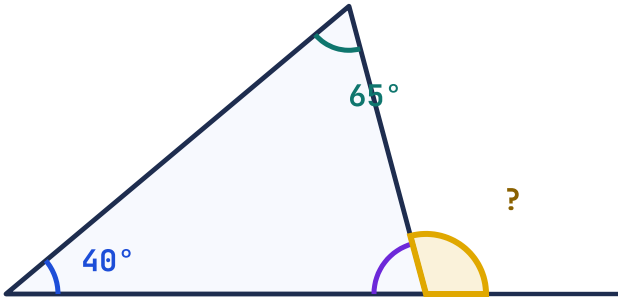
Exterior angle
the angle outside the triangle, next to one inside angle

Remote (far) interior angles
the two inside angles that do not touch the exterior angle

Watch out: Use the two **far** angles, not the angle right next to the exterior angle.



Walk through: The far inside angles are 40° and 65°. Find the exterior angle.



1. Which angles do we use?
   The two far inside angles: 40° and 65° → exterior = 40 + 65

2. Add them.
   105° → exterior = 105°

Check your understanding

1 Score 3 NC.8.G.5

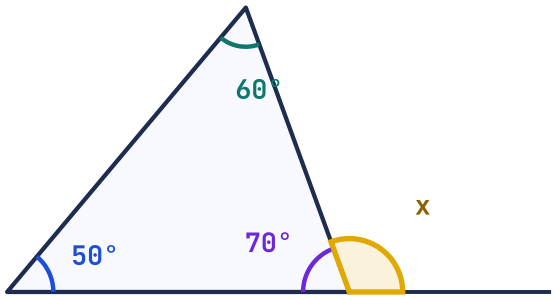
The far inside angles are 40° and 65°. Find the exterior angle.

Answer: **105°**

1. 40 + 65 = 105°

2 Score 3 NC.8.G.5

Find the exterior angle **x**.



Answer: **110°**

1. far angles: 50° and 60°
2. x = 50 + 60 = 110°